Exponential functions

1

a Yes

b No

2 A

3

a Decreasing
b Increasing
c Decreasing
d Increasing
e Increasing
f Increasing
g Decreasing
h Decreasing
i Decreasing
j Increasing
k Increasing
l Increasing

4 B

5

a $y = 5^x$

b $y = 3 \times 2^x$

6

a 1
b -2
c 3
d 3

7

a -243
b -3
c $-\frac{1}{3}$
d 4

8 67

9

a 1

b $\frac{9}{16}$

10 $50\,313.28$

11 6885.29

12 3313.46

13

a

i $\frac{1}{16}$ ii 1 iii 16

b None. For all values of x, it will have a positive value.

c It gets larger, approaching infinity.

d It gets smaller, approaching zero.

14

a

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y	$\frac{1}{243}$	$\frac{1}{81}$	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9	27	81	243	$59\,049$

b As x increases, the function increases at a faster and faster rate.

c All real x

d $y > 0$

15

a

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y	32	16	8	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{1024}$

b As x increases, the function decreases at a slower and slower rate.

c 1

d All real x

e $y > 0$

16

a

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y	$-\frac{1}{32}$	$-\frac{1}{16}$	$-\frac{1}{8}$	$-\frac{1}{4}$	$-\frac{1}{2}$	-1	-2	-4	-8	-16	-32	-1024

b No. For $a > 0$ the expression a^x is greater than zero for all real x. So, $-(a^x)$ will be less than or equal to zero.

c Decreasing

d As x increases, the function decreases at a faster and faster rate.

e All real x.

f $y < 0$

17

a Positive

b 0

c $y = 0$ is a horizontal asymptote of the curve.

18

a $(0, 1)$

b No

c All real x

d $y > 0$

e $y = 128$

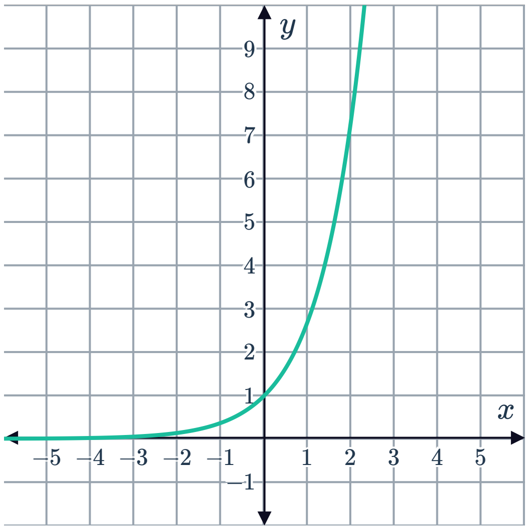
f $x = 8$

19

a

x	-3	-2	-1	0	1	2	3
$f(x)$	0.05	0.14	0.37	1.00	2.72	7.39	20.09

b



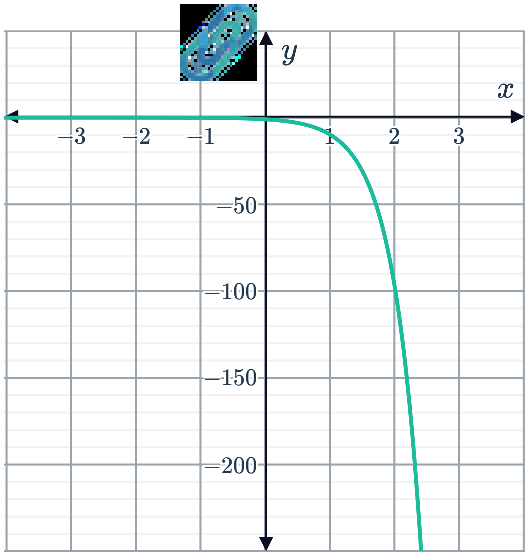
20

a

x	-2	-1	0	1	2	3
y	$-\frac{1}{100}$	$-\frac{1}{10}$	-1	-10	-100	-1000

b Yes. For any value of x, 10^x is always positive. Therefore, -10^x is always negative.

c



21

a No. For $a > 0$ the expression a^x is greater than zero for all real x.

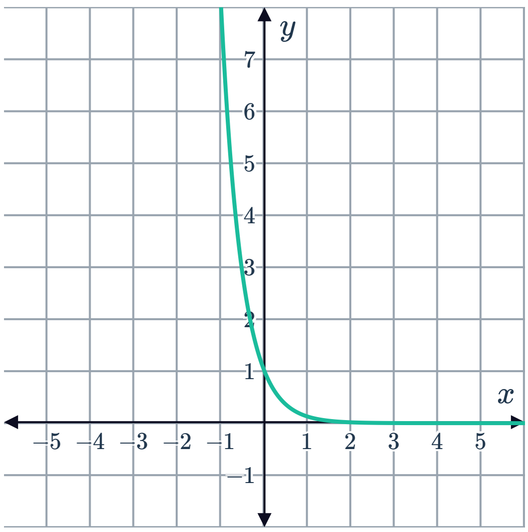
b 0

c ∞

d 1

e 0

f



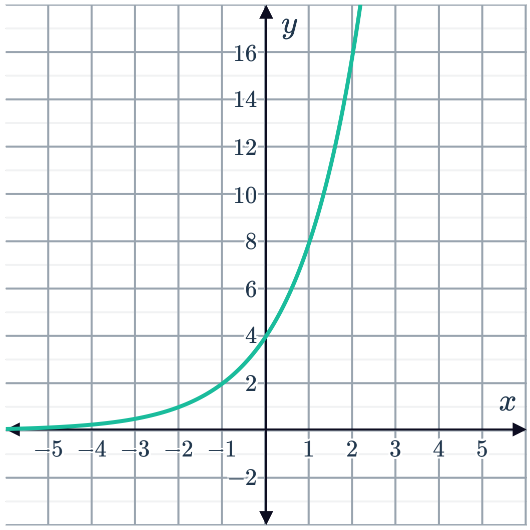
22

a $(0, 4)$

b No

c ∞

d



23

a 2

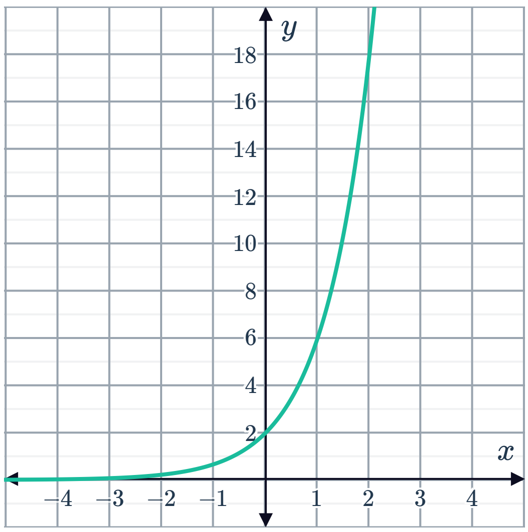
b No. For $a > 0$ the expression a^x is greater than zero for all real x. So, $2 \times (a^x)$ will be greater than or equal to zero.

c All real x.

d $y > 0$

e ∞

f



24

a Graph R

b For $x < 0$, the graph of $y = 6^x$ is below the graph of $y = 4^x$. This is because negative values of x result in fractional function values, and for any negative value of x, 6^x will result in a smaller value than 4^x.

25 a For $x > 0$, the graph of $y = e^x$ will lie above the graph of $y = (2)^x$ and below the graph of $y = (3)^x$.

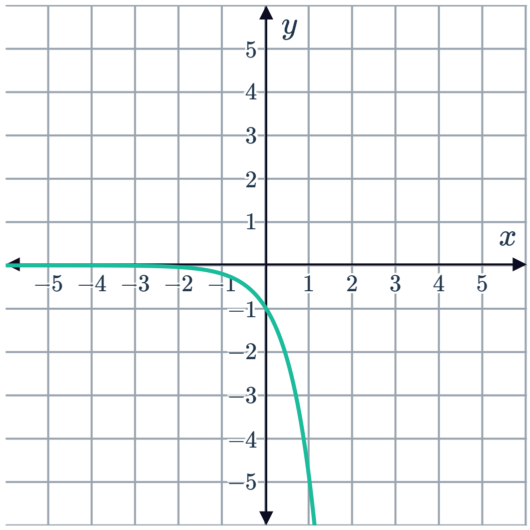
b For $x < 0$, the graph of $y = e^x$ will lie above the graph of $y = (3)^x$ and below the graph of $y = (2)^x$.

Transformations of exponential functions

26

a A reflection over the x-axis

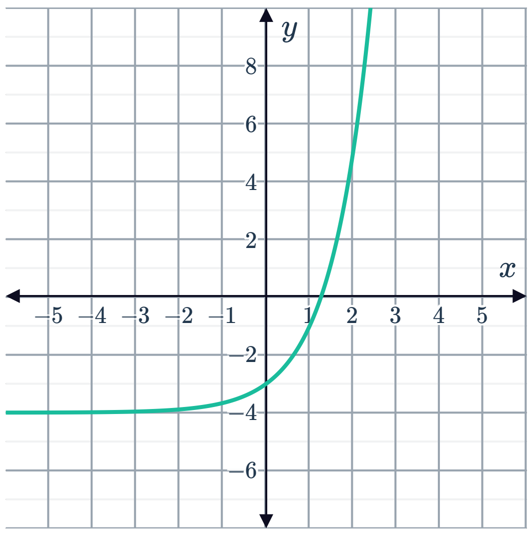
b



27

a Vertical translation

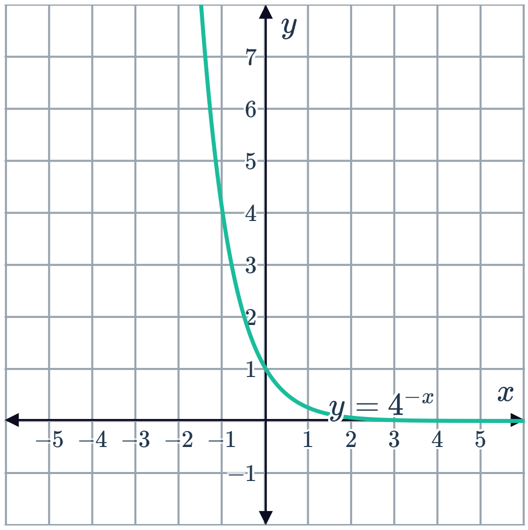
b



28

a   Reflection over the y-axis.

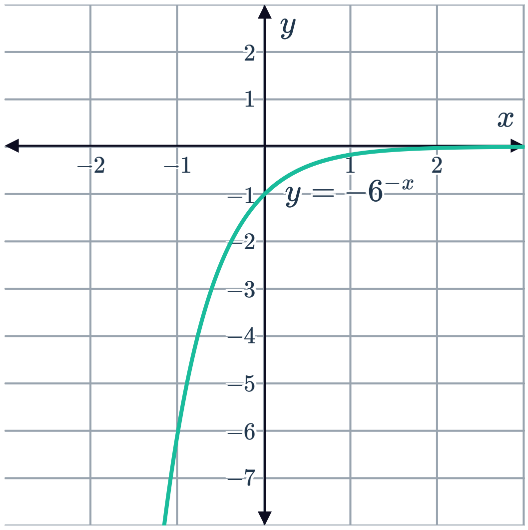
b



29

a   Reflection over the x-axis and y-axis.

b



30

a $y = 0$

b $y = 2$

c 1

d   All real x.

e $y < 2$

31

a 1

b -1

c $y = 0$

d $y = -2$

32

a

i 4 units to the left

ii 2 units

iii 1

b

i 1 units to the left

ii 1 units down

iii $\frac{1}{3}$

c

i $\frac{1}{5}$ units to the right

ii 2 units

iii 5

33

a $y = 2^x - 7$

b $y = 11^{x+10}$

34

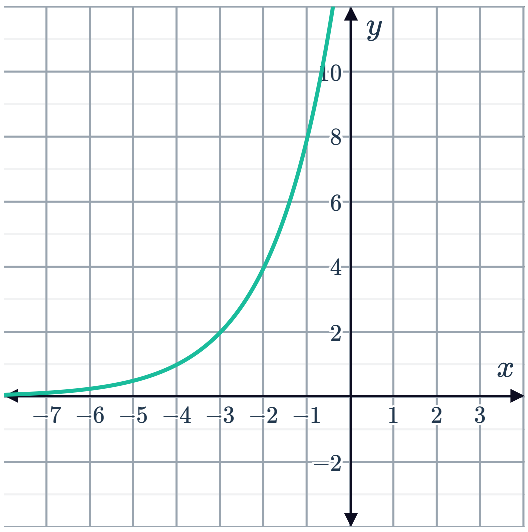
a $y = 7^{x-3} - 5$

b $y = -7^{x+3} + 5$

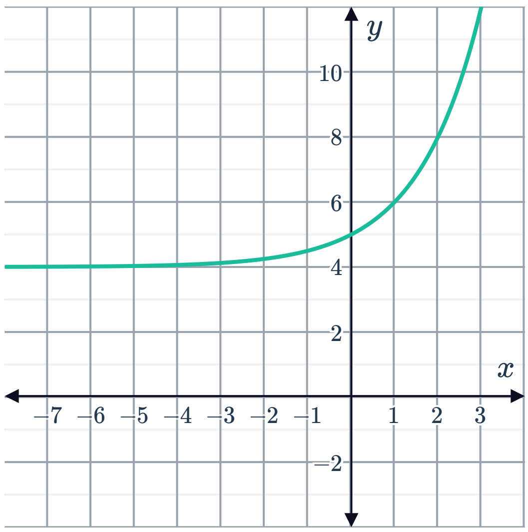
c $y = -7^{x+3} - 5$

35

a



b



36

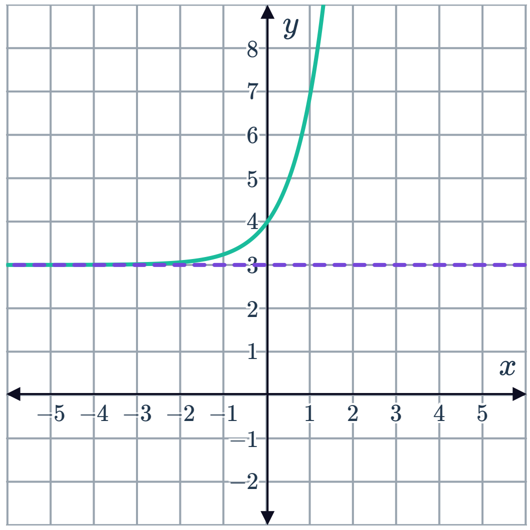
a

i 4

ii All real x

iii $y > 3$

iv



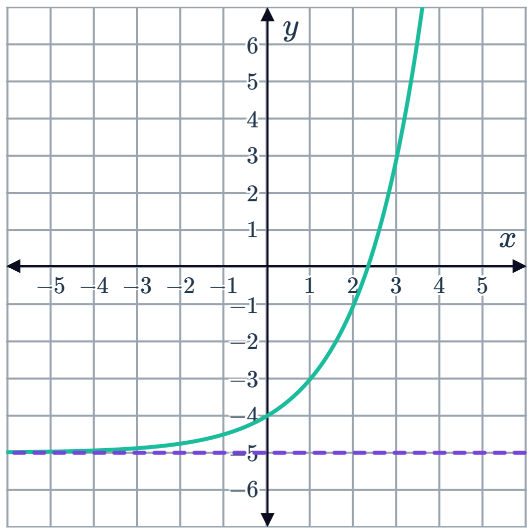
b

i -4

ii All real x

iii $y > -5$

iv



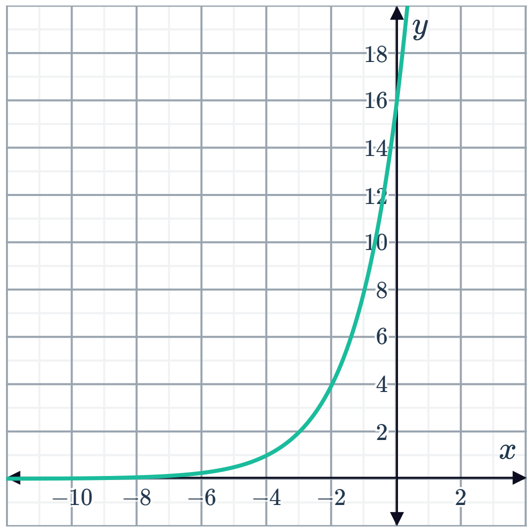
c

i 16

ii All real x

iii $y > 0$

iv



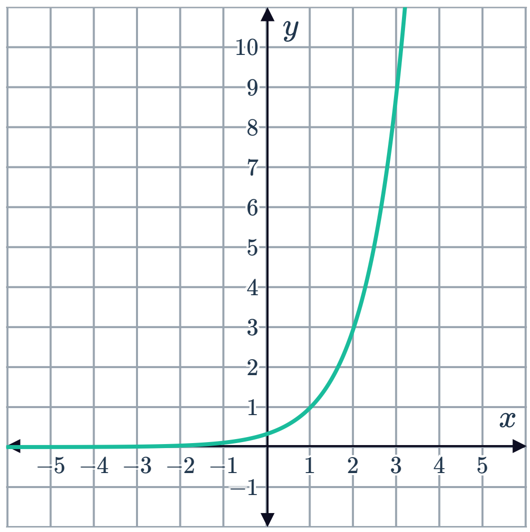
d

i $\frac{1}{3}$

ii All real x

iii $y > 0$

iv



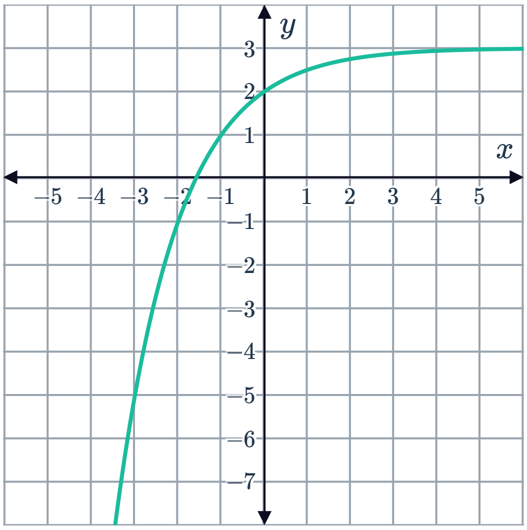
e

i 2

ii All real x

iii $y < 3$

iv



37

a $y = 0$

b $y = 2$

c 1

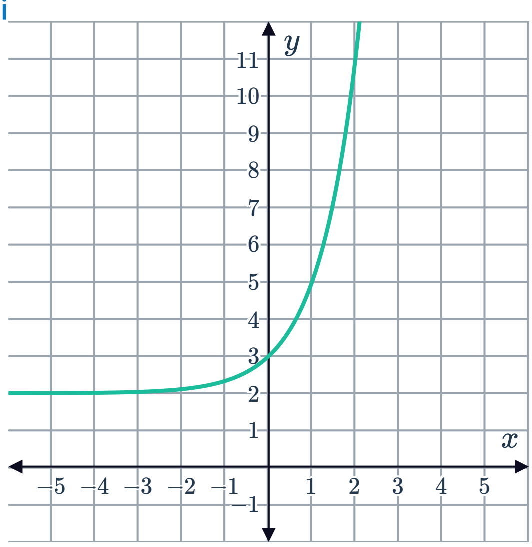
38

a

i $(0, 3)$

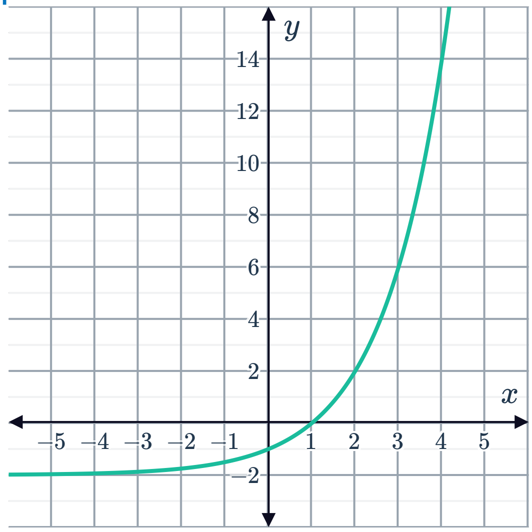
ii $y = 2$

iii



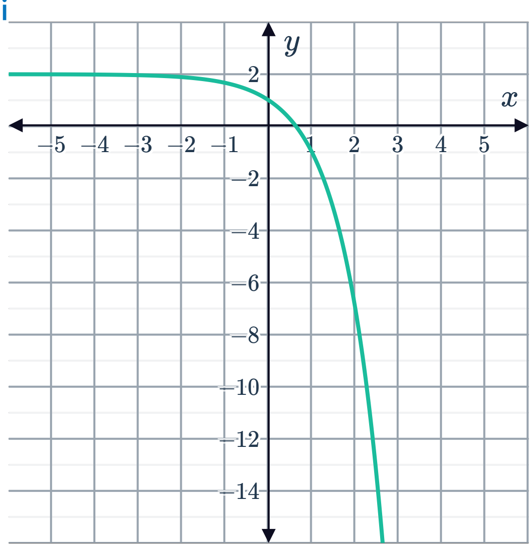
b

i $(0, -1)$ **ii** $y = -2$ **iii**



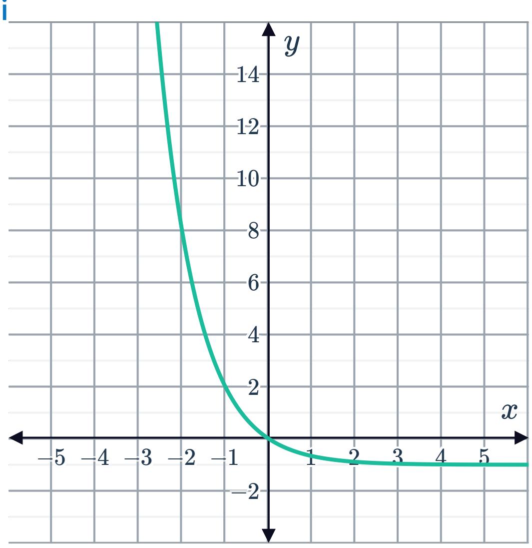
c

i $(0, 1)$ **ii** $y = 2$ **iii**



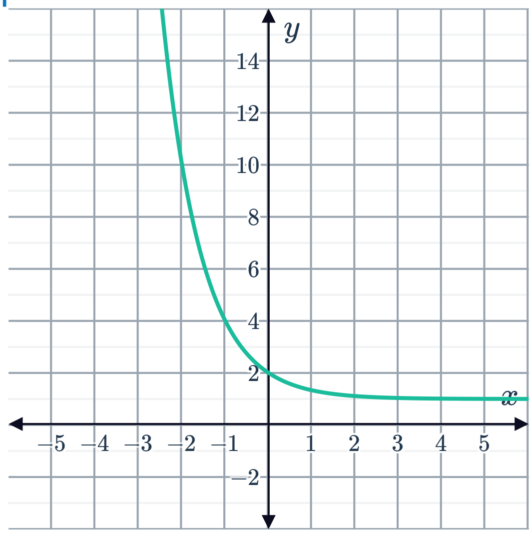
d

i $(0, 0)$ **ii** $y = -1$ **iii**



e

i $(0, 2)$ ii $y = 1$ iii



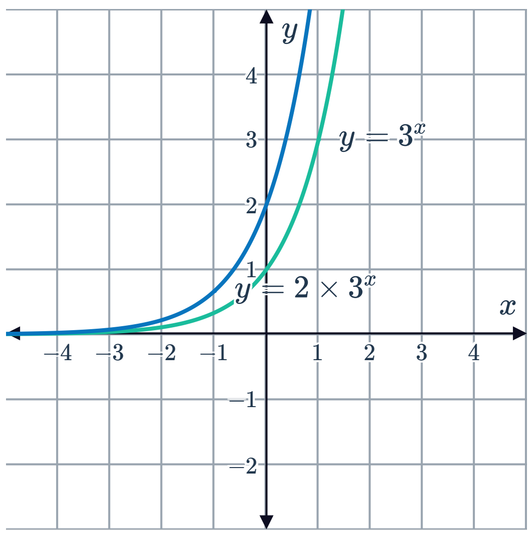
39

a

Point on original graph	Point on new graph
$\left(-1, \dfrac{1}{3}\right)$	$\left(-1, \dfrac{2}{3}\right)$
$(0, 1)$	$(0, 2)$
$(1, 3)$	$(1, 6)$
$(2, 9)$	$(2, 18)$

b $y = 2 \times 3^x$

c



d For all values of x, $y = 2 \times 3^x$ lies above $y = 3^x$.

40

a

i No ii No iii No

b

i No ii Yes iii Yes

c

i Yes ii Yes iii Yes

d

i No ii Yes iii Yes

Exponential equations

41

a $y = 2^x$ **b** $y = 2^{-x}$ **c** $y = 9^x$ **d** $a = 5$

42

a

x	1	2	3	4
y	2	4	8	16

b 2

c $y = 2^x$

43

a 10 **b** $y = 10^x$ **c** 10 000 000 000

44

a $a = 5$ **b** $b = 4$ **c** $y = \frac{5}{16}$